PCA

In [1]:
```
from sklearn.decomposition import PCA
from sklearn.datasets import load_iris

# Load dataset
X, y = load_iris(return_X_y=True)

# Create PCA model
pca = PCA(n_components=2)

# Transform data
X_pca = pca.fit_transform(X)

print("Shape after PCA:", X_pca.shape)
print(X_pca[:5])
```

```
Shape after PCA: (150, 2)
[[-2.68412563  0.31939725]
 [-2.71414169 -0.17700123]
 [-2.88899057 -0.14494943]
 [-2.74534286 -0.31829898]
 [-2.72871654  0.32675451]]
```

LDA

In [2]:
```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.datasets import load_iris

# Load dataset
X, y = load_iris(return_X_y=True)

# Create LDA model
lda = LinearDiscriminantAnalysis(n_components=2)

# Transform data
X_lda = lda.fit_transform(X, y)

print("Shape after LDA:", X_lda.shape)
print(X_lda[:5])
```

```
Shape after LDA: (150, 2)
[[ 8.06179978  0.30042062]
 [ 7.12868772 -0.78666043]
 [ 7.48982797 -0.26538449]
 [ 6.81320057 -0.67063107]
 [ 8.13230933  0.51446253]]
```

SVD

In [3]:
```
from sklearn.decomposition import TruncatedSVD
from sklearn.datasets import load_iris

# Load dataset
X, y = load_iris(return_X_y=True)

# Create SVD model
svd = TruncatedSVD(n_components=2)
```

```python
# Transform data
X_svd = svd.fit_transform(X)

print("Shape after SVD:", X_svd.shape)
print(X_svd[:5])
```

```
Shape after SVD: (150, 2)
[[ 5.91274714 -2.30203322]
 [ 5.57248242 -1.97182599]
 [ 5.44697714 -2.09520636]
 [ 5.43645948 -1.87038151]
 [ 5.87564494 -2.32829018]]
```